

PROBLEMS WITH CONCURRENT TRANSACTIONS

① Dirty Reads →

We might see uncommitted values in other concurrent transaction.

② Non Repeatable Reads →

In the same transaction, same SELECT query ran multiple times, might give different result

→ SELECT * FROM PRODUCT WHERE PRODUCT_ID = 1
① ↘ 20
 ↘ 10

③ Phantom Reads

If we insert new row into a table, we might see that coming in the other transaction.

Txn-1 : SELECT * - - -

Txn-2 : Insert a row.

Txn-1 : If we see the new row inserted that's a phantom read.

Phantom write

Txn 1

> SELECT * FROM PRODUCTS

1

2

...

...

10

Txn 2

> INSERT INTO PRODUCT
(11, "ABC" ...)

> COMMIT

> SELECT * FROM PRODUCTS

if we see a
new row
inserted → Phantom
Read

Avoided Phantom Read

1
2
...
10

SET 9 = 11
> [UPDATE PRODUCTS WHERE
PRODUCT ID = 11]

if this is successful
→ that means it is a phantom write

write is being performed on something that
was not visible

Indexing.

Purpose of Index:



To prevent unnecessary disk fetches
and hence optimise the retrieval



faster queries

Index → Hash Index

↳ not good for range
queries

↳ Everything in 1 table

B-Tree Index →

→ visualization

When should we create an index.

→ After creation of table

→ Judge on basis most frequent
queries that you get

Pro:

Reads for columns
on which index
has been created
are faster

Cons

writes are
slower
↓

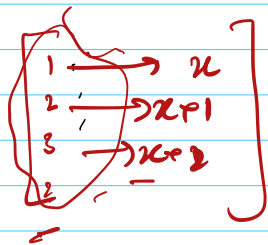
Overhead of
updating the

index

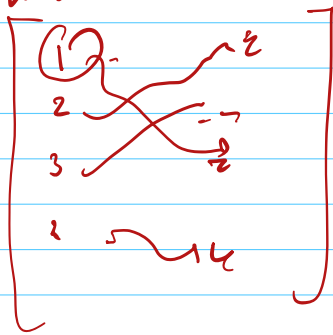
MySQL $\rightarrow$ We have records indexed implicitly on the basis of P.K

yes
clustered index

decide in what order records will actually be stored in HDD



quantity



$\rightarrow$ you create such index.

